

Revisiting Discrete Diffusion via Gaussian Duality: A Small-Scale Reproduction and Extensions of Duo

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Abstract—Uniform-state discrete diffusion models (USDMs) are appealing for fast text generation due to their parallel, iterative denoising and their ability to revise earlier decisions. However, they are commonly reported to lag behind autoregressive (AR) models and masked diffusion models (MDMs) on standard language modeling benchmarks. Duo (The Diffusion Duality) narrows this gap by establishing a theoretical connection in which a USDM can be viewed as emerging from an underlying Gaussian diffusion through a discretization operator, enabling the transfer of continuous-diffusion techniques to the discrete setting. In this work we (i) reproduce key findings of Duo at smaller scale under limited compute, and (ii) report additional experiments inspired by two complementary lines of work: Demystifying Diffusion Objectives, which motivates time-dependent reweighting of diffusion objectives as improved variational bounds, and MDLM frequency-informed training, which explores noise scheduling choices for masked diffusion language modeling. Our study aims to clarify what parts of Duo are most robust under compute constraints and how much headroom remains from objective reweighting and schedule design.

Keywords—Discrete Diffusion, Gaussian Diffusion, Loss Reweighting, Noise Scheduling

I. INTRODUCTION

Diffusion models have become a dominant paradigm in continuous domains, and discrete variants have been explored for language as a promising alternative to left-to-right autoregressive generation. Autoregressive (AR) models typically deliver strong likelihood-based performance but incur inherently sequential generation, which limits decoding parallelism and can amplify early mistakes. In contrast, discrete diffusion models generate through iterative refinement, which is naturally parallel over tokens and can self-correct by revising earlier tokens at later denoising steps. Within discrete diffusion, masked diffusion models (MDMs) corrupt tokens into a special mask symbol and learn to predict clean tokens; uniform-state models (USDMs) [4] instead move toward a simple categorical prior by progressively injecting uniform noise. Despite their conceptual appeal for fast generation, USDMs have historically been outperformed by AR and MDM baselines. Duo addresses this by showing that a uniform-state discrete diffusion process can be obtained from a Gaussian diffusion process [5] through discretization, and by leveraging this duality to improve both training and sampling [1]. This paper combines a compact exposition of diffusion for text with emphasis on Duo’s duality viewpoint, a sketch of the core proof idea and a small-scale reproduction plus extensions via objective weighting and noise schedule perturbations.

II. BACKGROUND

Let $x_{1:L}$ be a token sequence over vocabulary size K . Discrete diffusion introduces latent states z_t over timesteps $t \in [0, 1]$ that gradually transform the data distribution into a simple prior through a Markov forward process. In the interpolating noise view used by USDMs, the conditional marginal $q_t(z_t | x)$ is categorical and smoothly interpolates between the clean one-hot token at $t = 0$ and a near-uniform categorical distribution as $t \rightarrow 1$ [1].

A denoising model parameterized by θ is trained to approximate the reverse dynamics, enabling ancestral sampling from noise to data. Compared to AR decoding, diffusion offers parallel token updates; compared to MDMs, USDMs replace masking with uniformizing noise, which changes both the learning signal and the role of the noise schedule.

III. DUO AND THE DIFFUSION DUALITY

Duo’s key insight is that USDM trajectories can be seen as discretized Gaussian diffusion trajectories. Concretely, consider a Gaussian diffusion in continuous space producing latent vectors $w_t \in \mathbb{R}^K$ (per token position), and define the discretization operator

$$z_t = \operatorname{argmax}(w_t) \quad (1)$$

Duo shows that, under an appropriate correspondence between Gaussian diffusion coefficients $\bar{\alpha}_t$ and discrete coefficients α_t (via a transformation operator T), the discrete marginals induced by argmax match those of a valid uniform-state discrete diffusion process [1]. This provides a mechanism to transfer techniques from continuous diffusion (where design and training heuristics are more mature) to discrete diffusion for language.

A. Why this is attractive vs AR and MDMs

1) *Versus AR*: Diffusion enables parallel updates across positions and can revise past mistakes, whereas AR commits to a prefix and cannot edit earlier tokens during standard decoding.

2) *Versus MDMs*: MDMs rely on masking corruption and typically weight losses over noise levels. USDMs use a different corruption family (uniformizing categorical noise), and Duo’s duality offers principled ways to reuse continuous-diffusion recipes such as low-variance curricula and distillation strategies for few-step sampling [1].

IV. PROOF SKETCH OF THE DUALITY

In this section of the study, we outline the core argument at a high level.

A. Step 1: Pushforward of Gaussian marginals

Let $\bar{p}_t$ denote the Gaussian diffusion marginal over w_t . Define the pushforward distribution under discretization,

$$p_t = [\text{argmax}]_* \bar{p}_t \quad (2)$$

i.e., p_t is the distribution of $z_t = \text{argmax}(w_t)$ when $w_t \sim \bar{p}_t$. Duo proves that the Probability Mass Function (PMF) of z_t evolves according to an Ordinary Differential Equation (ODE) that uniquely characterizes a uniform-state discrete diffusion process, establishing that the discretized Gaussian trajectory corresponds to a valid USDM trajectory [1].

B. Step 2: Matching a USDM reverse kernel

Beyond matching marginals, one needs a reverse-time procedure implementable with a denoising model. Duo shows there exists a denoising model $x_\theta(z_t, t)$ such that the standard USDM ancestral sampler coincides with a marginal-preserving reverse process induced by the underlying Gaussian model. This guarantees that the USDM reverse-time chain maintains the same marginals as the discretized Gaussian diffusion across $t \in [0, 1]$ [1].

C. Step 3: Implications for likelihood bounds

An additional consequence highlighted by Duo is that the induced discrete likelihood can be at least as high as the Gaussian likelihood under the true data distribution (formalized as a theorem in the paper), reinforcing the appeal of the discrete process as more than a heuristic discretization.

V. TRAINING NOTES

Duo proposes a low-variance training strategy by relaxing the hard *argmax* discretization in the denoiser input to a tempered *softmax*,

$$\text{argmax}(w_t) = \lim_{\tau \rightarrow 0^+} \text{softmax}\left(\frac{w_t}{\tau}\right) \quad (3)$$

and training with $\tau > 0$ before annealing τ down, thereby reducing variance and controlling difficulty [1]. The duality transformation T also motivates curriculum choices over the Gaussian noise level $\bar{\alpha}_t$ that can cover a wide range of discrete noise strengths while sampling from a narrower (and potentially easier) interval in the Gaussian space.

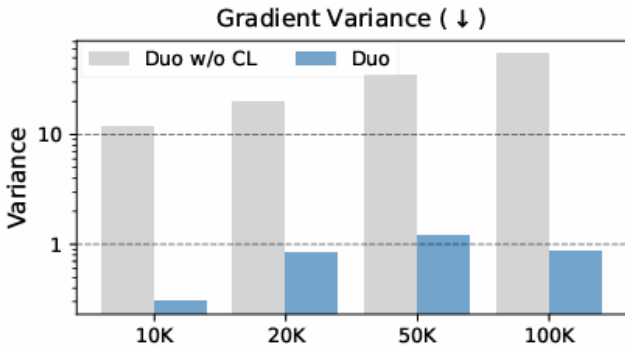


Figure 1 Curriculum learning drastically lowers the gradient variance in Duo trained with a fixed $\tau = 0.001$. The figure shows the summed gradient variance of the 100 weights with the highest variance, comparing Duo with CL (blue) and without CL (grey).

A. Curriculum Training

In standard discrete diffusion, during training we choose a random timestep, sample a noisy latent from the corrupted distribution and use a neural network to denoise it. We train the denoiser by minimizing the discrete NELBO which incorporates cross entropy terms and terms that account for the noise schedule. In the case of DUO, instead of sampling discrete latents, we sample gaussian latents and apply an *argmax* operation to go back to the discrete space—a method proven to be both equivalent and more efficient.

$$\begin{aligned} \mathcal{L}_{\text{train}} = \mathbb{E}_{x, t \sim \mathcal{U}[\beta, \gamma], \bar{q}t} \sum_{l \in [L]} f_{\text{Duo}}(z_t^l := \\ \text{argmax}(wt^l), x\theta([\text{softmax}(wt^{l'}/\tau)]_{l'=1}^L, t), a_t := \\ \mathcal{T}(\bar{\alpha}_t; x^l) \end{aligned} \quad (4)$$

However, the operator T that links the discrete with the gaussian noise schedule presents a challenge: even with minimal noise, a large vocabulary size makes it statistically likely for a non-target index to become dominant and through *argmax* it will be chosen. Therefore, we lose valuable signal during training which hinders the ability of the model to learn. To counter that the authors introduce *softmax* which is a relaxation of *argmax* and initially it allows the model to see more tokens and as the training progresses it eventually becomes an *argmax*. Furthermore, they do not choose a random timestep to sample but instead restrict the time in an interval where the signal changes significantly, thus providing the model with superior learning signal and stabilizing the training process as shown in Figure 1. Notably they also introduced a Rao-Blackwellized version of the loss function that seems to account significantly for the improved performance and quality of the model.

B. Sampling

To accelerate inference, DUO leverages the deterministic Gaussian ODE trajectory to perform knowledge distillation. A student model, initialized with the weights of the pre-trained teacher, is trained to emulate the teacher’s output. During this process, both models are presented with noisy latents; however, the teacher’s input is sampled from a point further along the trajectory (closer to the clean data). By minimizing the KL divergence between their outputs, the student learns to skip multiple refinement steps, ultimately producing high-quality samples with two orders of magnitude fewer steps as demonstrated in Figure 2. To further refine the results, DUO employs a greedy tail sampler in the final steps of inference to select the most probable tokens, while maintaining ancestral sampling in earlier stages to preserve generative diversity.

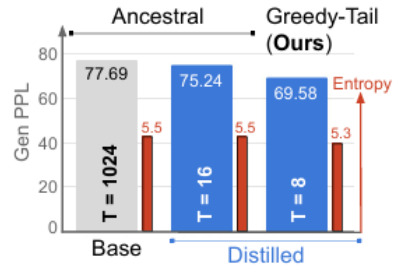


Figure 2 Sample quality comparison between the base Duo model and Duo distilled for 5 rounds using the DCD algorithm. The distilled model matches the base model’s sample quality in just 16 steps (vs 1024) with ancestral sampling. With the Greedy-Tail sampler, sampling steps can be further reduced to 8, achieving slightly better Gen PPL and lower entropy.

C. Proposed Improvements

In this work we first incorporate the reweighted loss framework proposed by Shi and Titsias [2]. Specifically, we employ three distinct weighting functions — Simple, FM and sigmoid — with the hyperparameter k set to 0. Secondly, we implement Frequency-Informed Training as suggested by Kosmopoulou et al. [3]. By computing the probability distribution of tokens within the vocabulary, we utilize a hyperparameter p to modulate the weight of each token. This enables a curriculum training strategy that prioritizes the learning of rare tokens during the initial stages of training.

VI. EXPERIMENTAL SETUP

This section details the data processing and computational configurations used to evaluate the proposed DUO variants against the established baselines. To ensure scientific rigor, all models, AR, MDML, DUO, and the four innovative variants, were trained under identical architectural and environmental conditions.

A. Dataset Construction and Pre-processing

For the training and validation purposes we utilized a subset of the OpenWebText (OWT) dataset. Given the 41GB and 4.5GB scale of the raw training and validation data respectively, which corresponds to 7.25M documents, and the limited computational resources, we developed a high-performance data engineering pipeline to create a 10,000-document training set and a 1,000-document validation set. The data engineering pipeline consists of three primary stages:

1) *Memory-Mapped Indexing*: To avoid the memory overhead of loading the entire corpus, we implemented a scanner using memory-mapped file I/O (mmap). The system identified the byte positions of every document occurrence. This generated a lightweight index allowing for instant access to any document in the file.

2) *Reproducible Randomised Sampling*: Using a fixed seed of 42, we performed randomized sampling from the index. We utilized byte-level seek operations to extract the selected 10,000 documents directly from the source dump, ensuring a zero-memory footprint during extraction.

3) *Multi-stage Text Sanitization*: The raw extraction contained significant non-linguistic noise. We applied a two-pass heuristic cleaning process:

a) *Artifact Removal*: Regular expressions were used to strip TAR headers (identified by the ustar signature) and null-byte padding.

b) *Metadata Stripping*: A secondary regular expression filter removed archival footers and hex-encoded document IDs.

Finally, the text was tokenized using the GPT-2 byte-level BPE tokenizer with a maximum sequence length of 512 tokens.

B. Model Architecture and Training Configuration

To train and evaluate the proposed variants, we employed a compact Discrete Diffusion Transformer (DDiT) architecture. The model consists of 8 Transformer blocks with 8 attention heads each, a hidden dimension of 256, and a condition dimension of 128. This configuration results in a

total of 33.8 million trainable parameters, representing an efficient "tiny" model scale.

All models were trained for 15,000 steps on NVIDIA T4 and P100 GPUs. We used a global batch size of 32 and the Adam optimizer. The learning rate was set to $5e-4$, with a 2,500-step linear warmup phase to ensure training stability. Training was performed using 32-bit precision (FP32) with an Exponential Moving Average (EMA) rate of 0.9999 for model weights.

C. Evaluation Benchmarks

Performance was quantified using the metrics Validation Perplexity, Bits Per Dimension, and Generated Perplexity, recorded every 1,000 steps. Upon completion of training, we conducted zero-shot evaluation to test the models' "out-of-distribution" robustness across four standard benchmarks. Those benchmarks are the datasets: LM1B, PTB, WikiText and LAMBADA.

VII. RESULTS AND ANALYSIS

In this part of the study, we provide a detailed analysis of the experimental results, focusing on the comparative performance of the baseline models and the impact of our proposed innovations.

We first evaluate the training efficiency and convergence patterns of all seven models across 15,000 steps. As illustrated in Figure 3, we successfully reproduced the performance hierarchy established in recent literature. The Autoregressive (AR) model exhibits a rapid plateau in Validation Perplexity and BPD at about 321 and 8.33 respectively, whereas the diffusion-based models (MDML and the DUO family) continue to improve throughout the 15,000-step window. Within the DUO family, the choice of weighting function $\hat{w}(t)$ proved critical. SIMPLE DUO performed the worst among the innovations, confirming that the more complex importance-sampling weights derived from Diffusion-Duality are necessary for linguistic modeling. Conversely, FM DUO and SIGMOID DUO achieved results nearly identical to the original DUO, with FM DUO achieving the most stable convergence in the final 5,000 steps. A striking disparity is observed in Generated Perplexity. While the AR baseline maintains high perplexity values of above 1000, the DUO variants consistently maintain values under 300. This indicates that DUO models are significantly more "confident" in their generations at small scales.

To test the models' ability to generalize beyond the training purposes, we conducted zero-shot evaluations. The results are summarized in Table 1. The zero-shot analysis reveals that MDML excels in three out of four datasets, whereas the DUO variants are superior in the LAMBADA benchmark. Specifically, standard DUO achieved the lowest perplexity at 727, indicating a stronger grasp of long-range dependencies and sentence-level logic. Our innovative FREQUENCY DUO produced interesting results on this scale, showing higher perplexity across the board. This suggests that while frequency-based scheduling might benefit larger models, at a 33M parameter scale, it introduces stochastic noise that hinders the model's ability to capture the fundamental "floor" of the language distribution.

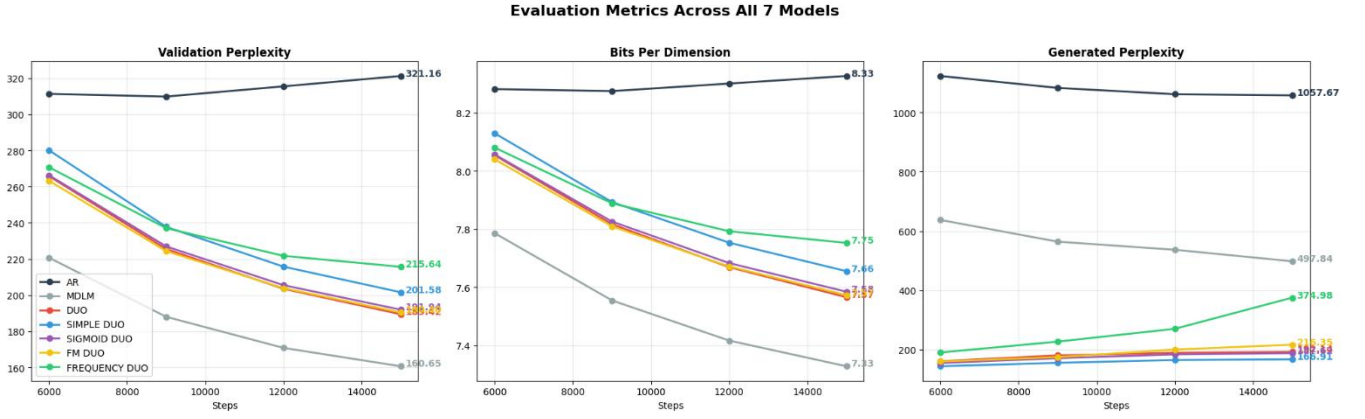


Figure 3 Comparative training dynamics across all models for Validation Perplexity, Bits Per Dimension (BPD), and Generated Perplexity.

TABLE I. ZERO-SHOT PERPLEXITY COMPARISON AT 15,000 STEPS

Perplexity(↓)	LM1B	PTB	WikiText	LAMBADA
AR	852	3081	3081	2237
MDML	384	1006	484	821
DUO	431	1130	682	727
SIMPLE DUO	474	1165	713	882
SIGMOID DUO	428	1458	786	794
FM DUO	476	1065	695	842
FREQUENCY DUO	475	1582	814	834

VIII. DISCUSSION

The experimental results provide several insights into the behavior of discrete diffusion models under constrained computational conditions.

The reduced-scale reproduction demonstrates that DUO’s architecture preserves stable convergence even when trained on a relatively small subset of OpenWebText (10,000 documents) and limited GPU resources. Validation perplexity and bits-per-dimension consistently improved throughout training, confirming that the diffusion objective remains well-behaved at smaller scale. Text generation quality improves respectively with training steps. Early-stage outputs (1k–3k steps) exhibit strong repetition and syntactic instability, which is expected during early diffusion model training. However, by 15k steps, generated text becomes structurally better and more comprehensible. This indicates that the model successfully learns meaningful token dependencies despite limited scale. Nevertheless, generated perplexity remains higher than validation perplexity, suggesting a discrepancy between likelihood optimization and sampling quality. This phenomenon is considered common in diffusion-based language models, where sampling complexity and multi-step denoising may introduce additional variance.

The investigation of alternative loss-weighting formulations confirms that time-dependent weighting plays a significant role in convergence behavior. Sigmoid-based and FM-inspired weighting schemes demonstrate slightly improved late-stage bits-per-dimension compared to the baseline DUO configuration. However, while convergence behavior improves modestly, the overall gains remain incremental rather than transformative. This suggests that weighting adjustments influence optimization efficiency more than model expressivity.

The frequency-informed scheduling strategy aimed to include token frequency information into the masking distribution. While validation perplexity and bits-per-dimension remain competitive, generated perplexity increases relative to the baseline DUO configuration. The detailed inspection of generated text indicates reduced coherence and increased repetition. This may suggest that frequency-based masking introduces bias toward high-frequency tokens, potentially weakening the model’s performance. Although these results highlight the sensitivity of diffusion-based language models to noise schedule design, it can be supported that a more carefully defined curriculum strategy could potentially improve overall robustness. This suggests that performance limitations observed in the frequency-informed variant may stem not from the concept itself, but from hyperparameter interaction between schedule design, curriculum progression, and objective reweighting.

IX. FUTURE WORK

Throughout this study, multiple directions for further research and improvement have been opened.

A. Scaling Data and Model Size

The experiments were conducted on a relatively small dataset subset due to computational limitations. Future work should explore training on larger datasets and increasing model capacity. Diffusion-based models may benefit significantly from scaling, as larger datasets could improve robustness across noise levels and enhance generation quality.

B. Increased Computational Resources

Diffusion models require multiple denoising steps during sampling, making them computationally more expensive than autoregressive models. Access to more powerful hardware would enable longer training schedules, which could allow the model to better converge and refine its representations across different noise levels. Increased computational capacity would also support larger batch sizes, deeper architectures, and a greater number of timesteps, all of which could contribute to improved model stability and generative quality. Furthermore, future work could explore sampling acceleration methods or step-reduction techniques to mitigate computational overhead and enhance the practical efficiency of diffusion-based language models.

C. Improved Noise Scheduling

The sensitivity observed in the frequency-informed scheduling experiments proves the significance of noise

schedule design in diffusion-based language modeling. Future research could investigate the development of learnable or adaptive noise schedules that dynamically adjust during training. Incorporating uncertainty-aware masking strategies may further improve robustness by allowing the model to focus on more informative denoising steps.

X. CONCLUSION

Overall, this work presented a reduced-scale reproduction and systematic analysis of the DUO diffusion-based language modeling framework. The study mainly aimed at the basic idea behind this model’s approach and evaluated the impact of alternative loss reweighting strategies and frequency-informed noise scheduling on training stability, convergence, and generation quality. The results demonstrate that DUO can be effectively reproduced under limited computational resources while maintaining stable optimization behavior. Moreover, some possible improvements were suggested, making us understand how loss reweighting influences convergence dynamics, and which role a noise scheduling plays in shaping model performance. Diffusion-based language models not only remain competitive, but they dominate autoregressive models in zero-shot perplexity under small-scale training conditions. These findings suggest that diffusion-based language modeling holds significant theoretical promise, but its full potential likely depends on larger-scale training and more refined scheduling strategies. Overall, this study contributes to a deeper understanding of how diffusion language models operate and how objective design and noise scheduling could influence the behavior of discrete diffusion language models and provides a foundation for future research in this evolving area.

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